Hypothesis Testing
Special Case of Two Groups

III. Multiple Comparisons (Isolating Differences)

A. Bonferroni t Test
B. Holm t Test

IV. Multiple Comparisons Against a Single Control

A. Bonferroni t Test
B. Holm t Test

V. The Meaning of P (read section in book, pgs. 67-71)

"The P value is the probability of obtaining a value of the test statistic as large or larger than the one computed from the data when in reality there is no difference between the different treatments"

ANOVA

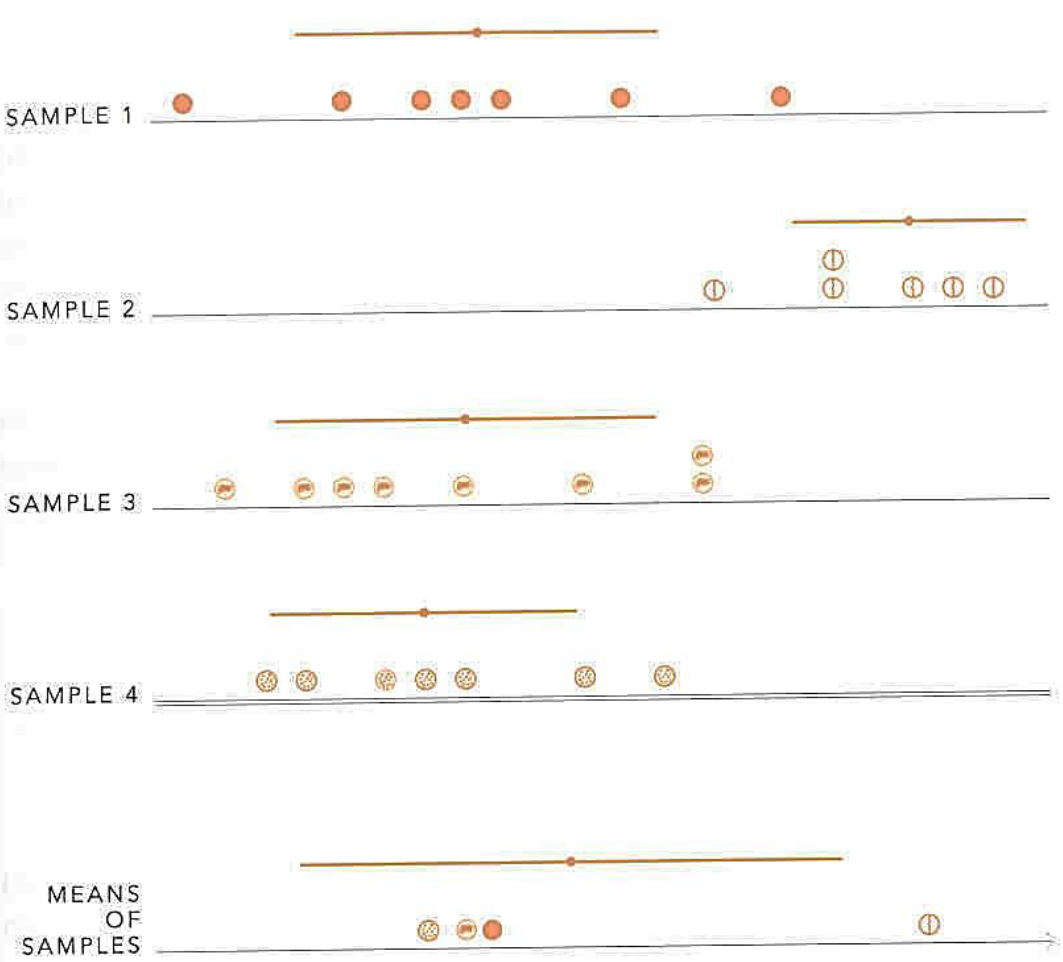


Figure 3–4 When the mean of even one of the samples (sample 2) differs substantially from the other samples, the variability computed from within the means is substantially larger than one would expect from examining the variability within the groups.

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_m$$

$$F = \frac{s^2_{bet}}{s^2_{wit}}$$

$$\nu_n = m - 1$$

$$\nu_d = m(n-1)$$

$$F > F_\alpha(\nu_n, \nu_d)$$

Reject H_0 with: $P < \alpha$

t Test

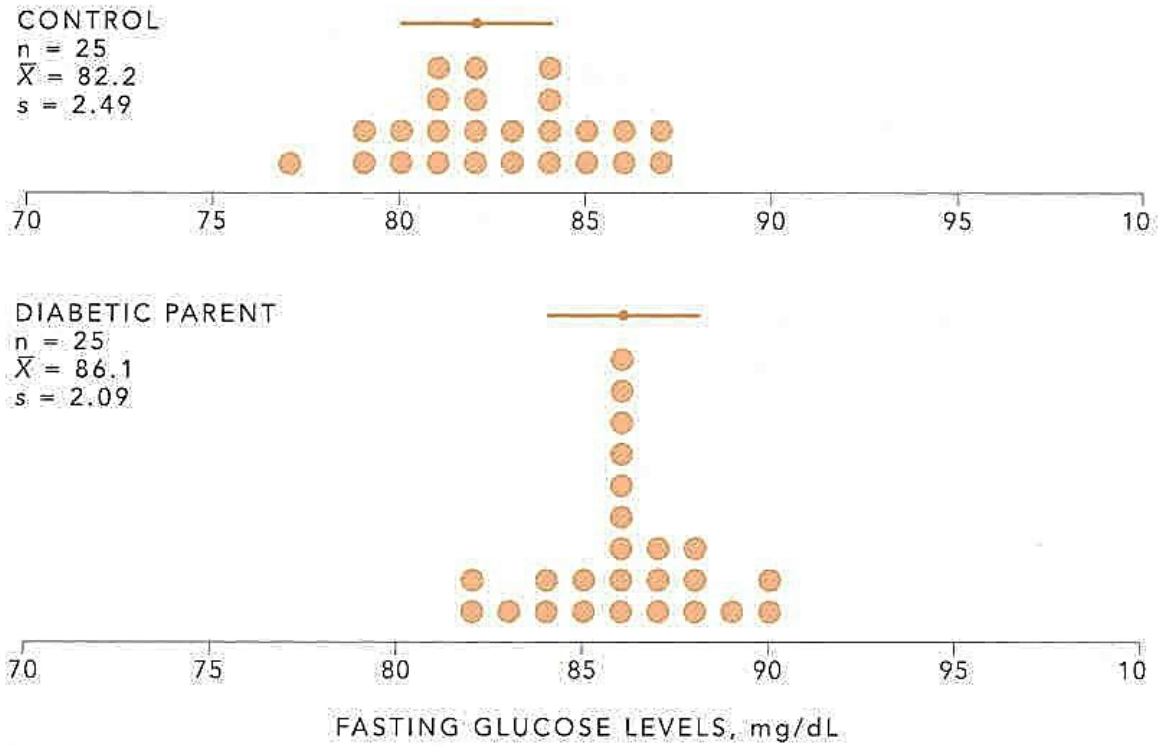


Figure 3–7 Results of a study comparing fasting glucose levels in children of parents who have type II diabetes and children of parents without type II diabetes. Each child's fasting glucose level is indicated by a circle at the appropriate glucose level. The average fasting glucose level for children of diabetic parents is higher than the average glucose level in children whose parents did not have diabetes. The statistical question is to assess whether or not the difference is due simply to random sampling or to an actual effect of the differences in parental history.

$$H_0 : \mu_1 = \mu_2$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}}$$

$$\nu = n_1 + n_2 - 2$$

$$|t| > t_\alpha(\nu)$$

Reject H_0 with: $P < \alpha$

Comments on Misuse of the *t* Test

Incorrectly-Applied Multiple-Comparison t Test

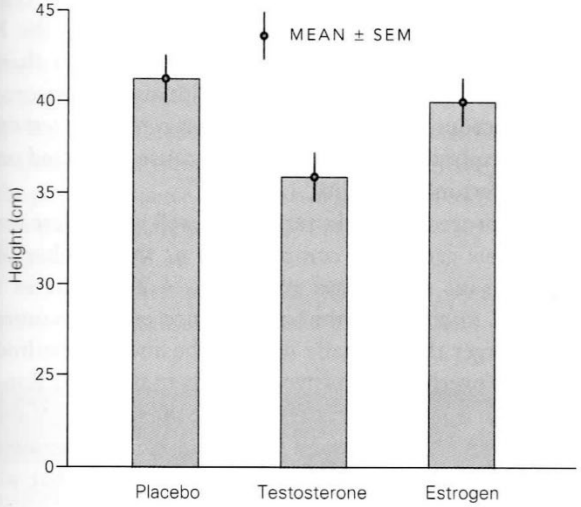


FIGURE 4-5. Results of a study of human hormones on Martians as it would be commonly presented in the medical literature. Each large bar has a height equal to the mean of the group; the small vertical bars indicate 1 standard error of the mean on either side of the mean (not 1 standard deviation).

(from Glantz, pg 61)

- 3 groups, 10 subjects per group
- Pairwise t Tests give the following results:
  - Placebo vs. Testosterone: $t = 2.39$
  - Placebo vs. Estrogen: $t = 0.93$
  - Testosterone vs. Estrogen: $t = 1.34$
- Degrees of freedom for each pair of groups: $\nu = n_1 + n_2 - 2 = 10 + 10 - 2 = 18$
- $t_{\text{crit}} = t_{0.05}(18) = 2.101$
- Results make no sense: PvsT different, but PvsE not different and TvsE not different:
  $\mu_P = \mu_E = \mu_T \neq \mu_P$

Isolating Differences Using t Test After Finding Difference Using ANOVA

Under H_0, best estimate of σ^2 is $s_{wit}{}^2$ from ANOVA (which was done first)

Degrees of freedom come with $s_{wit}{}^2$:

$$\nu = \nu_d = m(n-1) = N - m$$

$$t = \frac{\overline{x}_A - \overline{x}_B}{\sqrt{\frac{s_{wit}^2}{n_A} + \frac{s_{wit}^2}{n_B}}}$$

if n is same for all groups

more general formula

III A. Multiple Comparisons – Bonferroni *t* Test

Bonferroni inequality:

significance to use for each pairwise comparison

Overall level of Significance (e.g., 0.05)

$$\alpha_T < k\alpha$$

of pairwise comparisons

For each *t* Test comparison use:

$$\alpha = \frac{\alpha_T}{k}$$

if *n* is same for all groups

more general formula

Also $s^2_{wit} \rightarrow s^2$

$$\nu = \nu_d = m(n-1) = N - m$$

Role of Activity on HDL

$n = 70$	HDL Concentration (mg/dL)	
	Mean	Std. Dev.
(1) Marathon runners	64.8	14.3
(2) Joggers	58.0	17.7
(3) Inactive men	43.3	14.2

ANOVA Results:

$$F = \frac{s_{bet}^2}{s_{wit}^2} = \frac{8453.}{239.9} = 35.25$$

$$\nu_n = 3 - 1 = 2$$

$$\nu_d = 3(70 - 1) = 207$$

$$F = 35.25 > F_{0.01}(2, 207) = 4.60$$

Reject H_0 with: $P < 0.01$

Bonferroni *t* Test

$\alpha_T = 0.05$

$$\alpha = \frac{\alpha_T}{k}$$

	$\bar{x}_A - \bar{x}_B$	$\lvert t \rvert$	α	t_α
(1) vs (2) - runners vs joggers	6.8	2.6	0.017	2.41
(1) vs (3) - runners vs inactives	21.5	8.2	0.017	2.41
(2) vs (3) - joggers vs inactives	14.7	5.6	0.017	2.41

$$t = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{\frac{s^2_{wit}}{n_A} + \frac{s^2_{wit}}{n_B}}} \qquad s^2_{wit} = 239.9 \qquad \nu = \nu_d = 207$$

Reject H_0 that there is no difference in HDL levels among groups with $P < 0.01$

And conclude that HDL is different between runners and joggers
<u>and</u> between runners and inactives
<u>and</u> between joggers and inactives
} $P < 0.05$

III B. Multiple Comparisons – Holm *t* Test

Idea: Use successively larger significance levels

significance to use for the jth pairwise comparison

$j = 1,...,k$

Overall level of significance (e.g., 0.05)

$$\alpha_j = \frac{\alpha_T}{(k - j + 1)}$$

of pairwise comparisons

For example:

$k = 3$

$\alpha_1 = 0.0167$ largest value of $|t|$

$\alpha_2 = 0.025$:

$\alpha_3 = 0.05$ smallest value of $|t|$

Role of Activity on HDL

$n = 70$	HDL Concentration (mg/dL)	
	Mean	Std. Dev.
(1) Marathon runners	64.8	14.3
(2) Joggers	58.0	17.7
(3) Inactive men	43.3	14.2

ANOVA Results:

$$F = \frac{s_{bet}^2}{s_{wit}^2} = \frac{8453.}{239.9} = 35.25$$

$$\nu_n = 3 - 1 = 2$$

$$\nu_d = 3(70 - 1) = 207$$

$$F = 35.25 > F_{0.01}(2, 207) = 4.60$$

Reject H_0 with: $P < 0.01$

Holm *t* Test

$\alpha_T = 0.05$

$$\alpha_j = \frac{\alpha_T}{(k - j + 1)}$$

	$\bar{x}_A - \bar{x}_B$	$\lvert t \rvert$	α	t_α
(1) vs (3) - runners vs inactives	21.5	8.2	0.017	2.41
(2) vs (3) - joggers vs inactives	14.7	5.6	0.025	2.26
(1) vs (2) - runners vs joggers	6.8	2.6	0.05	1.97

$$t = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{\frac{s^2_{wit}}{n_A} + \frac{s^2_{wit}}{n_B}}} \qquad s^2_{wit} = 239.9 \qquad \nu = \nu_d = 207$$

Reject H_0 that there is no difference in HDL levels among groups with $P < 0.01$

And conclude that HDL is different between runners and joggers
and between runners and inactives
and between joggers and inactives
} $P < 0.05$

IV A. Multiple Comparisons Against Control
Bonferroni *t* Test

Bonferroni inequality:

significance to use for each pairwise comparison

Overall level of Significance (e.g., 0.05)

$$\alpha_T < k\alpha$$

of pairwise comparisons

For each *t* Test comparison use:

$$\alpha = \frac{\alpha_T}{k}$$

if *n* is same for all groups

more general formula

Also $s^2_{wit} \rightarrow s^2$

$$\nu = \nu_d = m(n-1) = N - m$$

Bonferroni *t* Test Example

$$\alpha = \frac{\alpha_T}{k} \qquad \alpha_T = 0.05$$

	$\bar{x}_A - \bar{x}_B$	$\lvert t \rvert$	α	t_α
(1) vs (3) - runners vs inactives	21.5	8.2	0.025	2.26
(2) vs (3) - joggers vs inactives	14.7	5.6	0.025	2.26

$$t = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{\frac{s_{wit}^2}{n_A} + \frac{s_{wit}^2}{n_B}}} \qquad s_{wit}^2 = 239.9 \qquad \nu = \nu_d = 207$$

Reject H_0 that there is no difference in HDL levels among groups with $P < 0.01$

And conclude that HDL is different between runners and inactives
and between joggers and inactives
} $P < 0.05$

IV B. Multiple Comparisons Against Control
Holm t Test

$$\alpha_j = \frac{\alpha_T}{(k-j+1)} \qquad \alpha_T = 0.05$$

	$\bar{x}_A - \bar{x}_B$	$\lvert t \rvert$	α	t_α
(1) vs (3) - runners vs inactives	21.5	8.2	0.025	2.26
(2) vs (3) - joggers vs inactives	14.7	5.6	0.05	1.97

$$t = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{\frac{s^2_{wit}}{n_A} + \frac{s^2_{wit}}{n_B}}} \qquad s^2_{wit} = 239.9 \qquad \nu = \nu_d = 207$$

Reject H_0 that there is no difference in HDL levels among groups with $P < 0.01$

And conclude that HDL is different between runners and inactives
<u>and</u> between joggers and inactives
} $P < 0.05$

t Test Functions in R

From the "lsr" package

- ***independentSamplesTTest()***: performs two-sample *t* Test (i.e., *t* Test on data from two groups made up of different individuals)
- ***oneSampleTTest()***: performs single-sample inference against a given reference value
- ***posthocPairwiseT()***: performs pairwise comparisons using results from ***aov()***

Generic function

- ***t.test()***: can perform many types of *t* Tests, depending on the specific arguments entered

Session 9 Lecture Exercise

An analysis resulted in five pairwise comparisons and, therefore, five calculated t values (see table below). The ANOVA done prior to performing these pairwise comparisons had $\nu_n = 3$ and $\nu_d = 46$. Use the Holm correction to identify which pairwise comparisons indicate a significant difference between groups and which do not.

Comparison	***t* value**
G1 vs G2	2.1
G1 vs G3	2.9
G1 vs G4	2.2
G1 vs G5	2.3
G1 vs G6	3.3